

Human-Computer Interaction

Generative AI, Creativity & Tools for Thought

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Today's Agenda

- Topic overview: *Generative AI, Creativity & Tools for Thought*
- The frame: *does the tool augment the thinker, or substitute for the thinking?*
- Provocation: *Does GenAI grow or diminish human thinking?*
- Discussion

This term's frame

Enduring fundamental → How AI is changing it → Depth (a claim to argue with)

- **Fundamental:** computing as an *intellectual amplifier*, extending thought (Bush, Engelbart)
- **How AI changes it:** generation makes the **artifact** cheap; the scarce skills become *steering* and *verification*
- **Depth: Deep:** it questions what the tool is for, and starts the *deskilling* thread

This week's readings

- **Required:** Lee et al., *The Impact of Generative AI on Critical Thinking* (CHI 2025)
- **Recommended:** Bush, *As We May Think* (1945), re-read from Week 3
- **Optional:** Dhillon et al., *Varied Scaffolding Levels in Co-writing with Language Models* (CHI 2024)

Topic overview: ***Generative AI, Creativity & Tools for Thought***

The lineage: tools for thought

Two ideas, both about making the human more capable:

Externalize (*Bush, 1945*)

Store and link thought outside the head: "an enlarged intimate supplement to his memory"

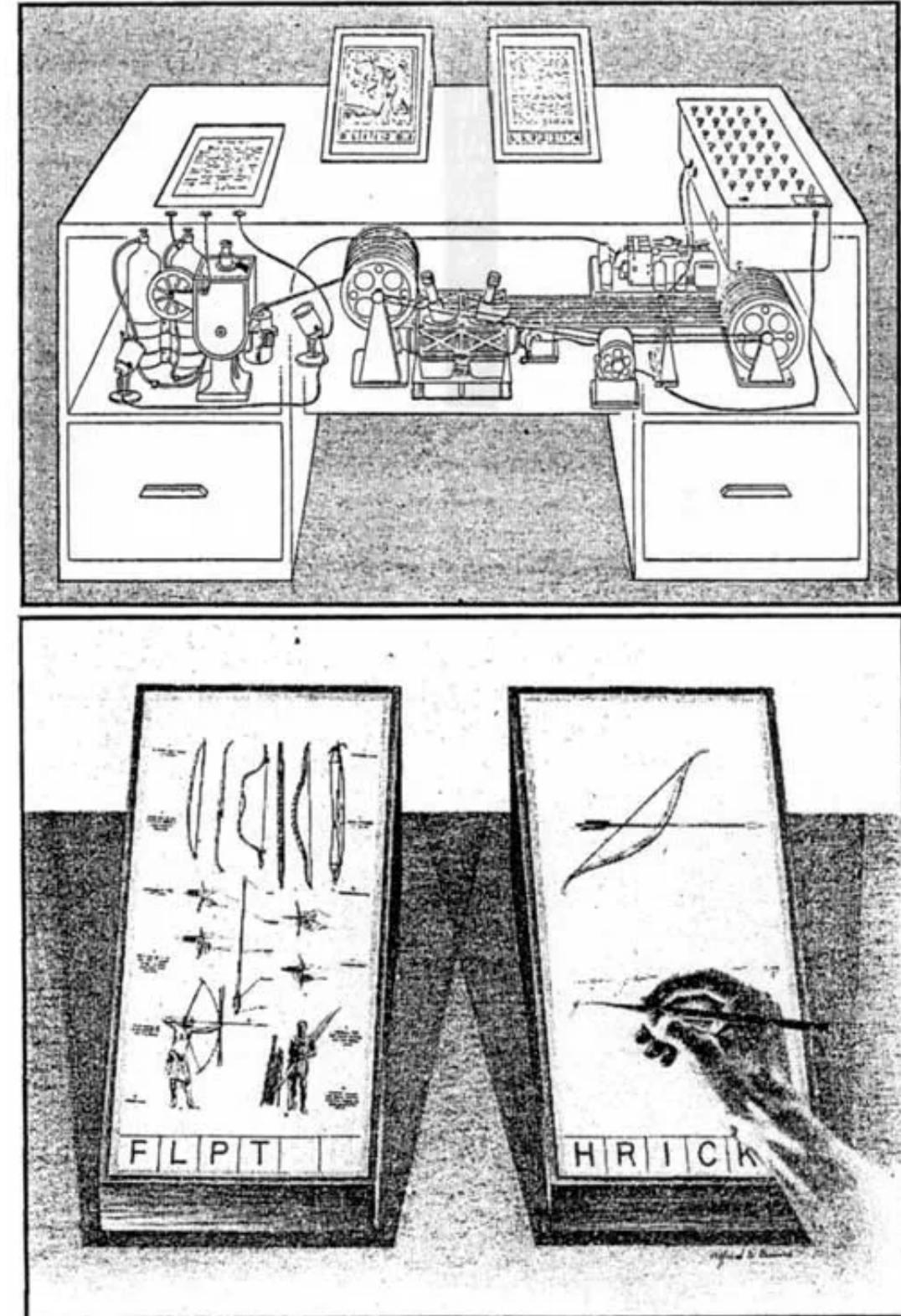
Augment (*Engelbart, 1962*)

"increasing the capability of a man to approach a complex problem situation"¹

¹ Engelbart, D. (1962). *Augmenting Human Intellect: A Conceptual Framework*. SRI Summary Report AFOSR-3223, §1a.

1945: Bush's memex²

- A desk that stores "all his books, records, and communications"
- Organized the way the mind works: it "operates by association"
- The line he drew: "For mature thought there is **no mechanical substitute**."



² Bush, V. (1945). [As We May Think](#). *The Atlantic*.

Distributed cognition³

Thinking isn't confined to "the skin or skull of an individual." It is spread:

- **across people**: a crew, a team
 - **between inside and outside**: notes, instruments, tools
 - **through time**: earlier work reshapes later work
- 👉 External aids don't just add memory; they **change the task**.

³ Hollan, Hutchins & Kirsh (2000), *ACM TOCHI*, 7(2), 174-196.

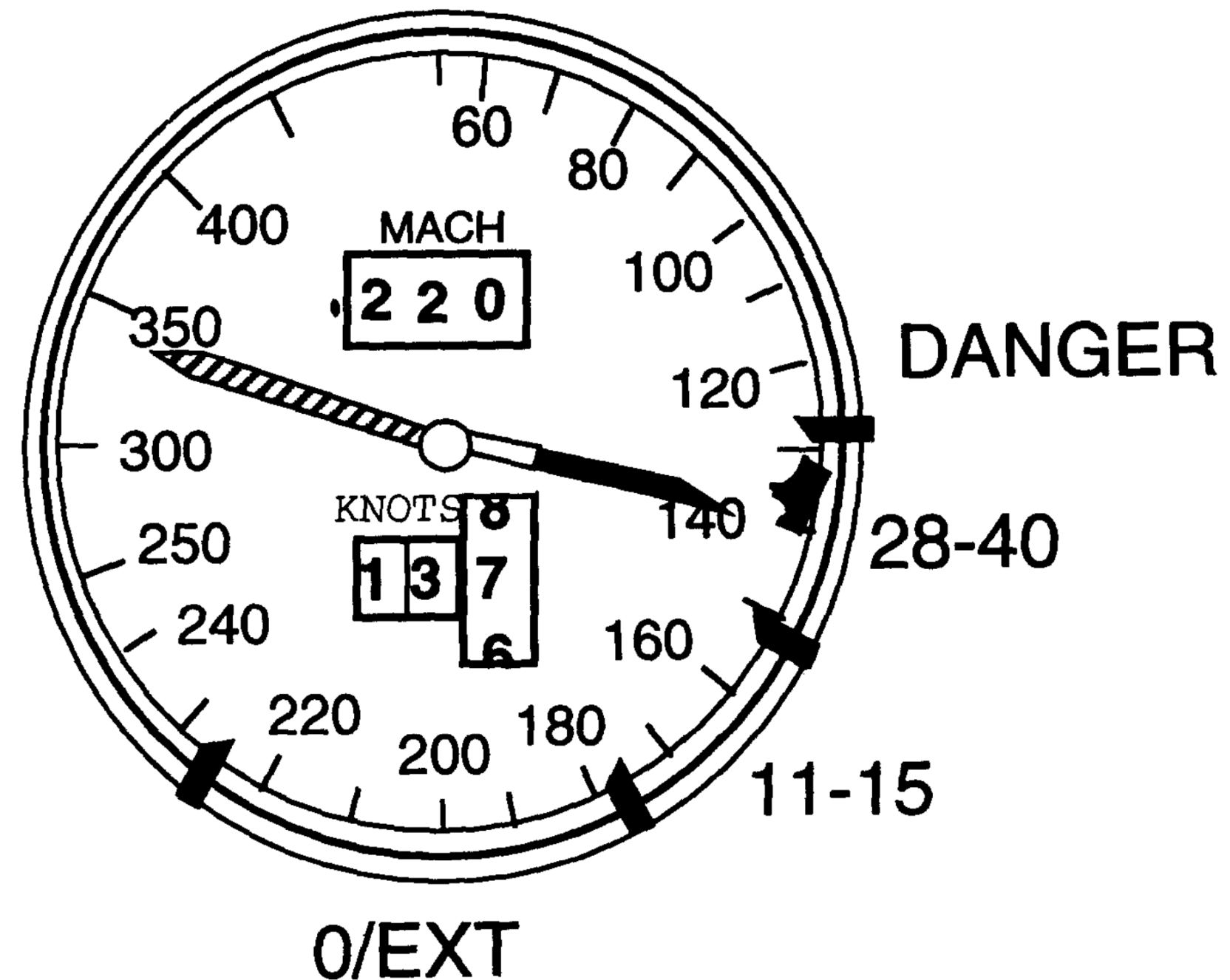
How a cockpit remembers its speeds⁴

Pilots set **speed bugs** on the airspeed dial before landing.

- Worked out early, read later at a glance
- Memory becomes perception: is the needle at the bug?

"Speed bugs do not help pilots remember speeds; rather, they are part of the process by which the cockpit system remembers speeds."

⁴Hutchins (1995), *Cognitive Science*, 19(3). Fig. 4.



What AI changes: the cost of the artifact

Engelbart's 1962 "writing machine" made drafts cheap, so the writer could think more.

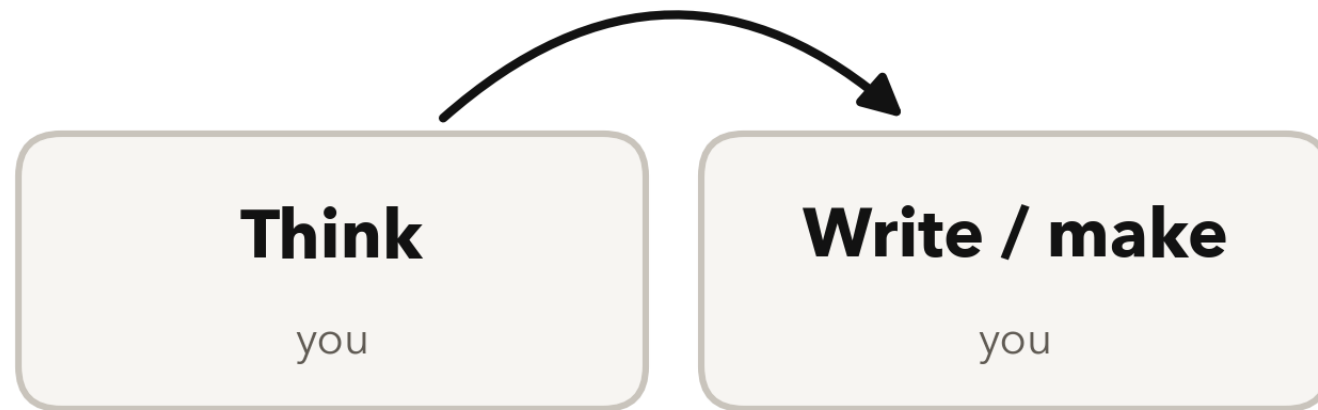
Generative AI makes the artifact cheap by **writing it for you**.

- The bottleneck moves from *making the thing* to *knowing what to make, and whether it's good*
- New core skills: **steering** (what to ask for, keep, and change) and **verification**

👉 Cheap output is not the same as understanding.

The old loop

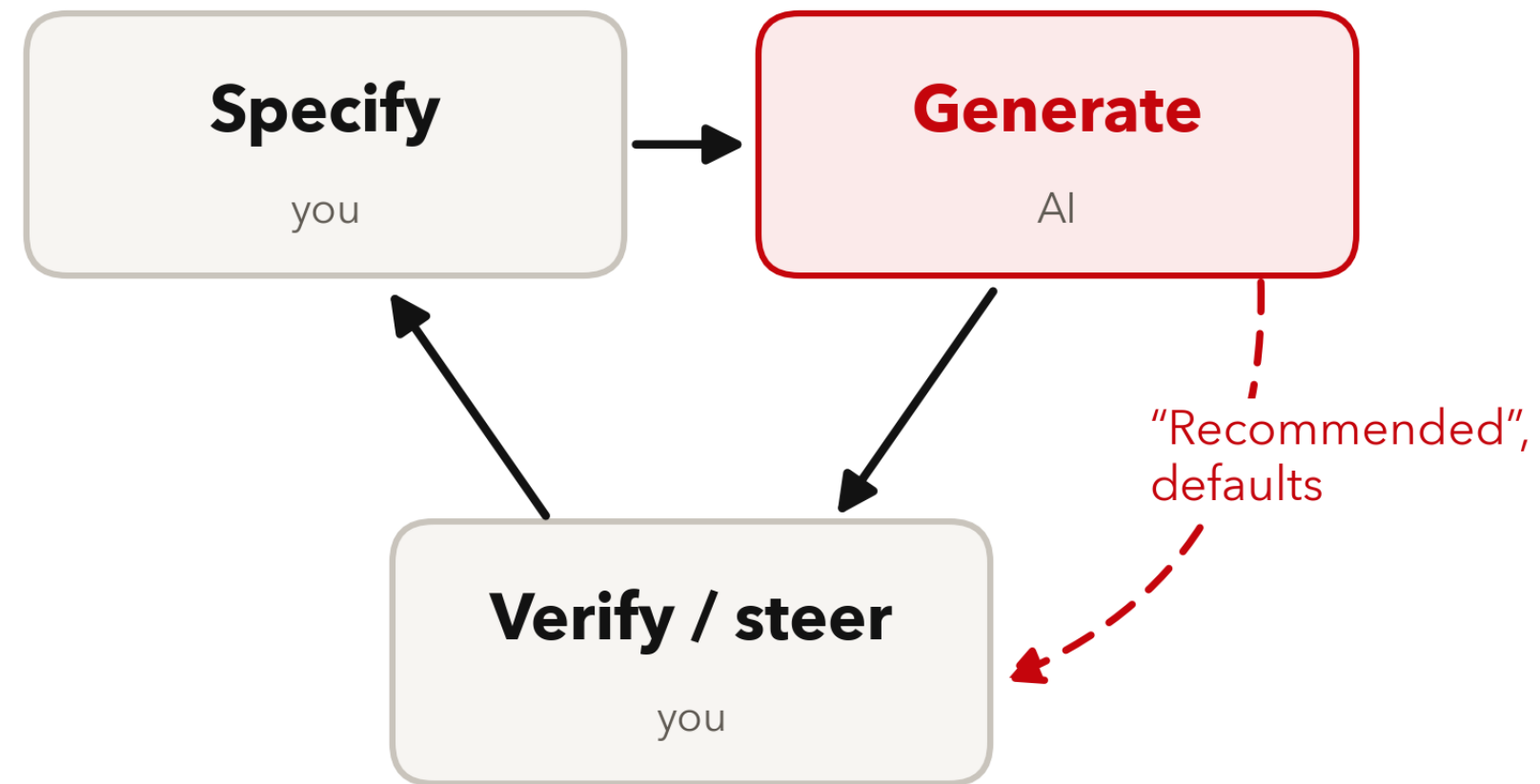
think and make, tightly coupled



Judgment is exercised
on every turn

The new loop

specify, generate, verify, repeat



the tool reaches into the judgment, too

The old loop is where judgment develops. The new loop spends it.

Where does judgment come from?

Our claim: the old loop **builds** the judgment the new loop **spends**.

- **Generate it, remember it**⁵
 - **Math students:** better with GPT-4, then 17% worse without it⁶
 - **Struggling novice coders:** "an illusion of competence"⁷
- 👉 Professionals spend judgment. Students are still building it.

⁵ Slamecka & Graf (1978), *J. Exp. Psych.*

⁶ Bastani et al. (2025), *PNAS*.

⁷ Prather et al. (2024), *ICER*.

It grades its own options, too

A "**Recommended**" tag hands over the other half: *whether it's good*.

- **Defaults stick:** the preselected option tends to win⁸
- **Suggestions narrow:** shorter, more predictable writing⁹
- **Suggestions persuade:** users' own views shifted¹⁰
- **Advice gets followed,** even when it's wrong¹¹

⁸ Johnson & Goldstein (2003), *Science*.

⁹ Arnold, Chauncey & Gajos (2020), *IUI '20*.

¹⁰ Jakesch et al. (2023), *CHI '23*.

¹¹ Parasuraman & Manzey (2010), *Human Factors*.

Designing against it

- **Decide first, then see the AI:** cognitive forcing reduced overreliance, though people liked it least¹²
- **AI that challenges, not obeys:** a "Socratic gadfly" that questions your choices instead of making them¹³

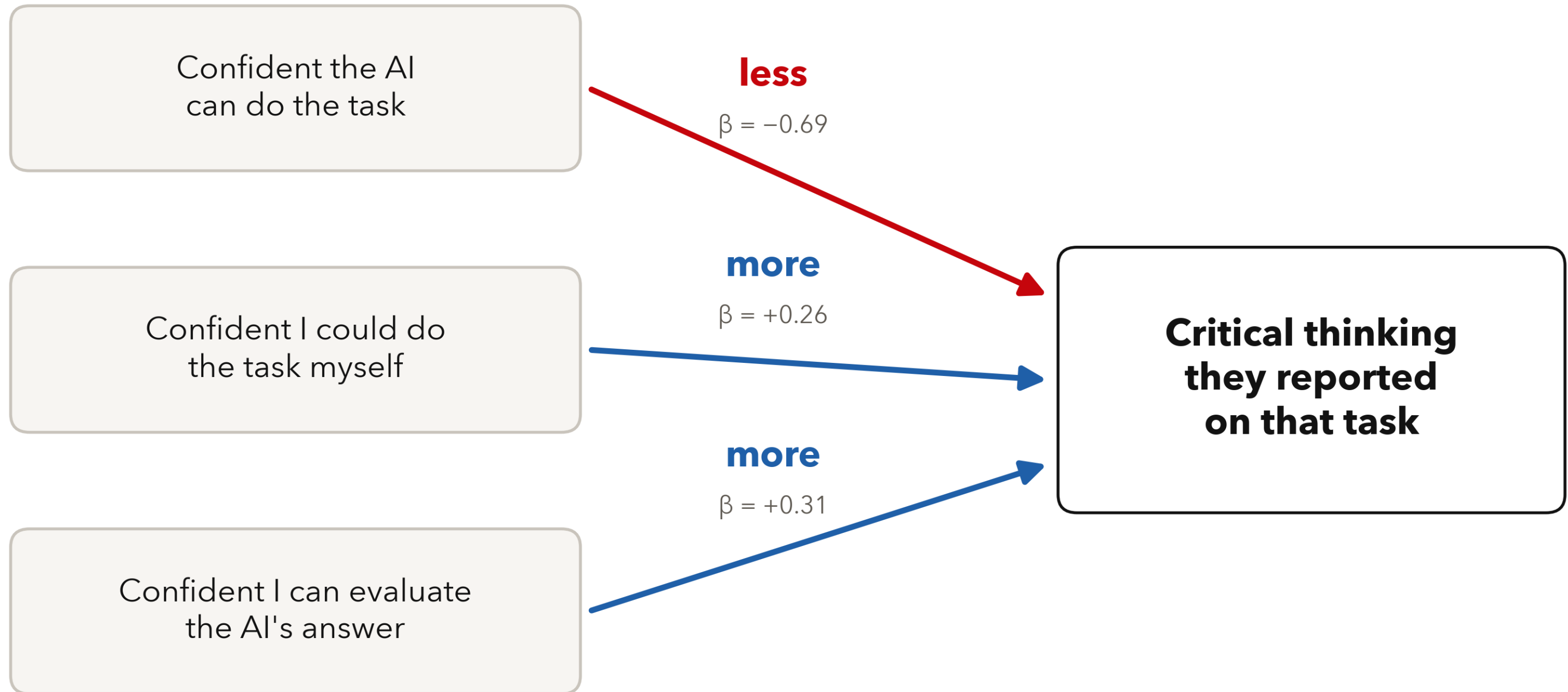
👉 Design test: does the tool leave room for *your* judgment before it offers its own?

¹² Buçinca, Malaya & Gajos (2021), CSCW (Week 4).

¹³ Sarkar, A. (2024). AI Should Challenge, Not Obey. *CACM*, 67(10), 18-21.

Confidence and critical thinking

Lee et al. (2025): 319 knowledge workers, 936 real uses of GenAI at work



Each task rated by the person who did it. All self-reported; correlations, not causes.

Lee et al. (2025), CHI '25, §4.2.1: mixed-effects regression on perceived enactment of critical thinking.

The central tension

Bush drew the line: "For mature thought there is no mechanical substitute."

Augment

The tool takes the repetitive part,
and you end up **more capable**
(Engelbart)

Substitute

The tool takes the thinking itself;
without practice, the skill can
erode (Bainbridge, via Lee et al.)

How much should the tool write?¹⁴

131 people wrote with no AI, with next-sentence, or with next-paragraph suggestions.

- **Paragraphs:** better quality and productivity, most for less regular writers
- **Sentences:** no significant gain
- **Both:** lower ownership and satisfaction; same workload

👉 Not "less is better": the help that raised quality lowered ownership.

¹⁴Dhillon, P.S., Molaei, S., et al. (2024). Shaping Human-AI Collaboration. *CHI '24*. doi:10.1145/3613904.3642134

Can you verify what the tool shows you?

Lee et al.: with GenAI, critical thinking shifts toward **verification**.

- Tools label their answers: high, medium, or low confidence
- But a label is itself a claim, and it can be wrong
- One answer can't tell you: a "medium" answer is *supposed* to be wrong sometimes

👉 The only check is the tally: of all the "medium" answers, how many were right?

You can only check a label in aggregate

40 answers from an AI tool, tallied by confidence label:

Label	Claims	Actually right
High	85% or more	18 of 20 (90%)
Medium	70-84%	4 of 10 (40%)
Low	under 70%	6 of 10 (60%)

👉 "Medium" was wrong most of the time; "Low" did better.

Guardrails coda

Three questions to carry into any creativity tool:

- **Attribution / authorship**: who is the author when the artifact is generated?
- **Over-reliance**: does fluent output inflate confidence past competence?
- **Whose labor**: whose work did the corpus *launder* to make this cheap?¹⁵

¹⁵ Baio, A. (2022). [AI Data Laundering](#). Waxy.org.

**But what about the
thinker?**

The dialectic, today

Generative AI is the newest move in the tools-for-thought lineage.

- **Bush, Engelbart:** augment the mind, don't spare it
 - **Lee et al.:** trust in the tool, less reported critical thinking
 - **Dhillon et al.:** more help, better text, less ownership
- 👉 Design test: are you a **better thinker** when it's off?

Your provocation (*post one slide*)

Pick one:

1. **Provocation**: does generative AI *grow* or *diminish* human thinking? Ground it in the readings.
2. **Critical artifact**: a real GenAI tool that makes the *steering* / *authorship* / *deskilling* tension visible
3. **Design / policy**: one guideline that keeps the human the *thinker*, not the spectator

Discussion

Today's roles

Reporter

Skeptic

Connector

Seat 4

Seat 5

Seat 6

Same in every pod. **Seats 1, 2 and 3** have no role today; you are still in the argument.

Discussion Format

In your pod, on your pod monitor: ~30 minutes.

- **Everyone shows their slide:** about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines (agreed, split, unresolved) and delivers them to the room

What's Next?

- **Friday: Experimental Design + Measurement (M5)**
 - **Read:** LFH Ch. 2-3
 - Post your team's 3-line project status
- **Due Fri Oct 9, 9:30 am:** Qualitative Field-Method Plan
(individual)